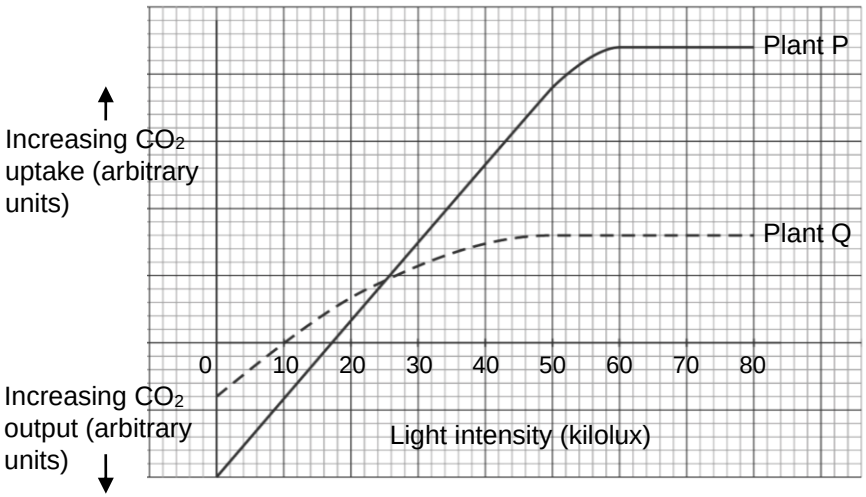
Multiple-choice questions

|!|BCEMA0100000002|!| ★★

The graph below shows the carbon dioxide uptake and release of two different plant species at different light intensities.



With reference to the graph, which of the following combinations is correct?

	Type of plant	***Light intensity at which the compensation point is reached***
A.	Plant P is a sun plant.	60 kilolux
B.	Plant P is a shade plant.	17 kilolux
C.	Plant Q is a sun plant.	50 kilolux
D.	Plant Q is a shade plant.	10 kilolux

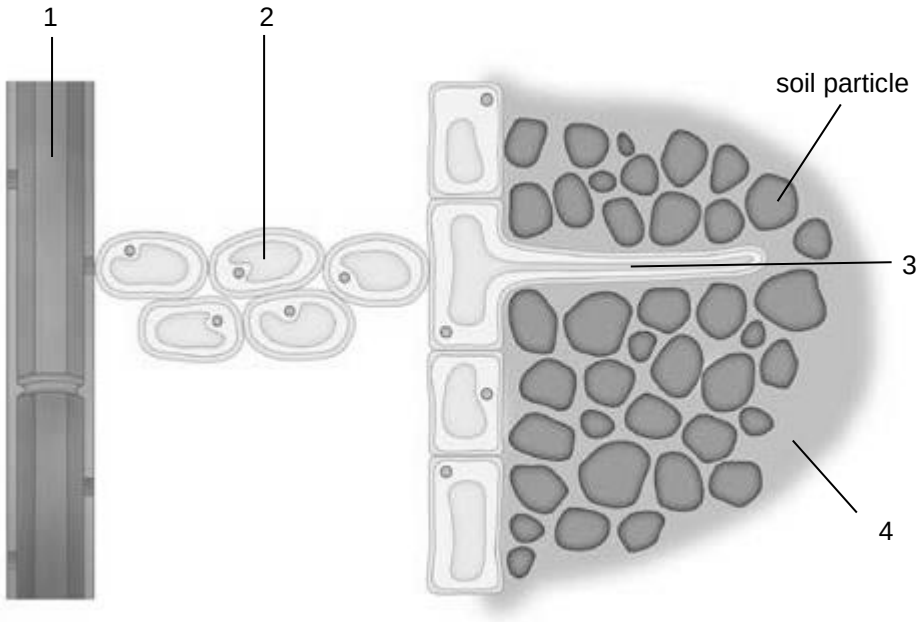
##D##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0100000003|!| ★

The diagram below shows the transverse section of a certain part of a young root in the soil.



Which of the following correctly arranges the water potential of different root cells and soil water in descending order?

A. 2, 3, 4
B. 2, 4, 3
C. 3, 4, 2
D. 4, 3, 2

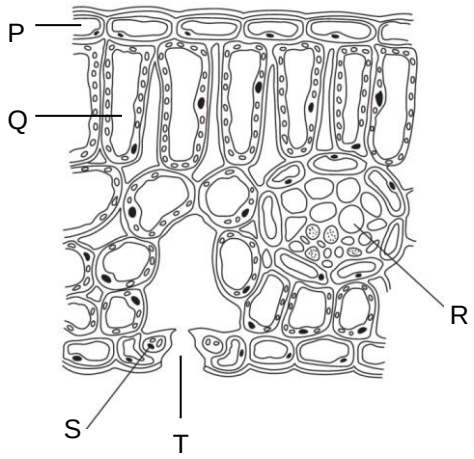
##D##

Skill assessed	Mark
Understanding basic concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0100000004|!| ★

The diagram below shows the transverse section of a leaf.



Between which two points does most gas exchange take place?

A. P and Q
B. P and S
C. Q and R
D. Q and T

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0100000005|!|

Which of the following is ***not*** the result of nitrogen deficiency in flowering plants?

A. curling of leaves
B. slow root growth
C. slow shoot growth
D. yellowing of leaves

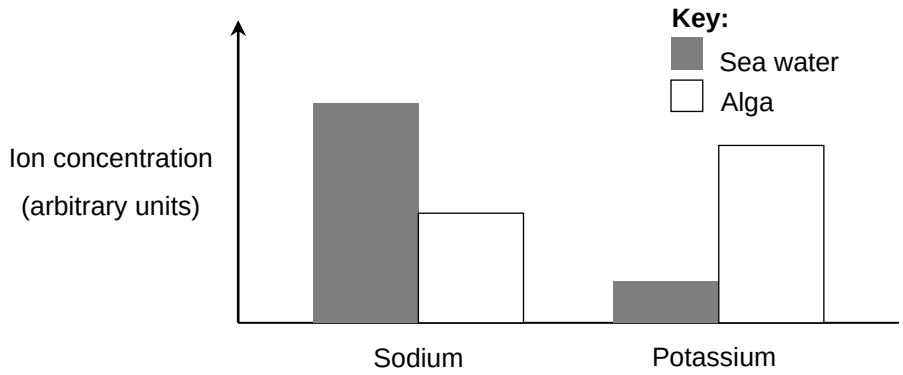
##A##

Skill assessed	Mark
Understanding basic concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMC0100000006|!|

The bar chart below shows the concentration of ions in a single-celled alga and the surrounding sea water.



Which of the following statements is correct?

A. Sodium ions will move out of the alga by diffusion.
B. Sodium ions will move into of the alga by active transport.
C. Potassium ions will move out of the alga by diffusion.
D. Potassium ions will move into the alga by active transport.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0100000007|!| ★ **SI**

A small aquatic plant was put in each of three Petri dishes, X, Y and Z, with different culture solutions. The table below shows the experimental results after six weeks.

Petri dish	Results
X	The plant had more leaves than it had previously, and they are of normal size but very pale.
Y	The plant had the same number of leaves as it had previously and they were all small and yellowish.
Z	The plant had more leaves than those in X and Y, and they were of normal size and much darker in colour.

Which of the following shows the elements that were missing from the culture solutions?

	X	***Y***	***Z***
A.	magnesium	nitrogen	phosphorus
B.	magnesium	phosphorus	nitrogen
C.	nitrogen	phosphorus	magnesium
D.	phosphorus	magnesium	nitrogen

##A##

© 2019 Aristo Educational Press Ltd.

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0100000008|!|

Which of the following elements is required by plants in the least amount for normal growth?

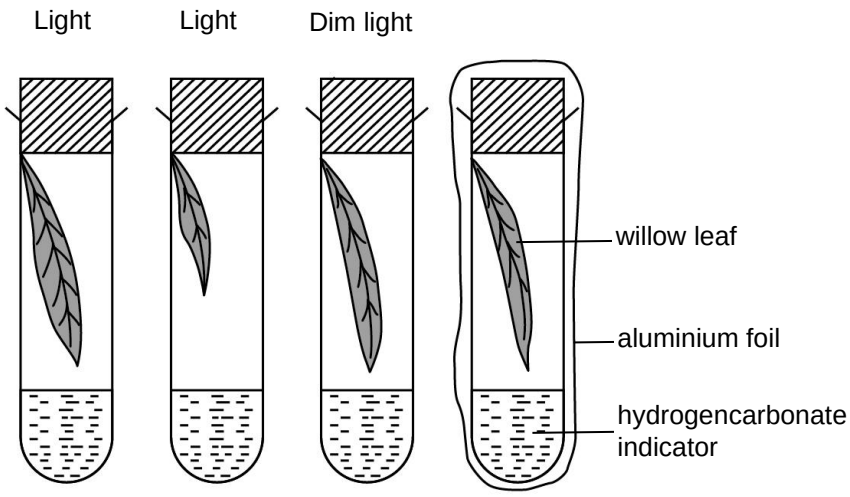
A. phosphorus
B. potassium
C. copper
D. magnesium

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0100000009|!| ★ **SI**

Equal volume of hydrogencarbonate indicator solution is added to each of the four boiling tubes. A willow leaf is also added to each of them.



In which tube will the indicator solution changes from red to purple first?

A. (1)
B. (2)
C. (3)
D. (4)

##A##

Skill assessed	Mark
Understanding basic concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0100000011|!| ★

The table below shows the uptake of some nutrients through the roots of a plant in the presence and absence of oxygen.

Nutrients	Rate of uptake (arbitrary units / day)	
	Presence of oxygen	Absence of oxygen
Water	39.2	39.0
Nitrates	21.2	1.7
Phosphates	15.1	2.2

What of the following conclusions can be drawn from the data in the table?

A. Photosynthesis can speed up the absorption of nitrates and phosphates from the soil.
B. The uptake of nitrates can occur when the concentration of nitrates in the root hair cells is higher than that in the soil water.
C. The uptake of phosphates cannot take place in the absence of oxygen.
D. The uptake of water requires oxygen.

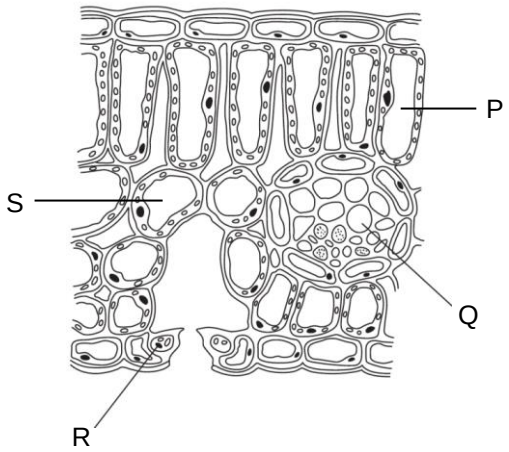
□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0100000012|!| ★

The diagram below shows the transverse section of a leaf.



Which of the following cells has the highest rate of carbon dioxide uptake during daytime?

A. cell P
B. cell Q
C. cell R
D. cell S

□

##A##

© 2019 Aristo Educational Press Ltd.

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0100000014|!|
Which of the following substances are obtained from the soil for plants to carry out photosynthesis?

(1) carbon dioxide
(2) water
(3) magnesium

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0100000015|!|
Which of the following combinations about the functions of nitrogen and phosphorus in plants is ***incorrect***?

	Nitrogen	***Phosphorus***
A.	as the component of amino acids	as the component of nucleic acids
B.	as the component of nucleic acids	as the component of amino acids
C.	as the component of chlorophyll	as the component of phospholipids
D.	as the component of proteins	as the component of ATP

##B##

Skill assessed	Mark
Understanding basic concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMC0100000016|!|

Which of the following elements is/are required by plants for making chlorophyll?

(1) nitrogen
(2) phosphorus
(3) magnesium

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0100000017|!|

Which of the following parts of a terrestrial dicotyledonous plant is/are ***not*** covered by cuticle?

(1) the upper epidermis of a leaf
(2) the lower epidermis of a leaf
(3) the epidermis of a root

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

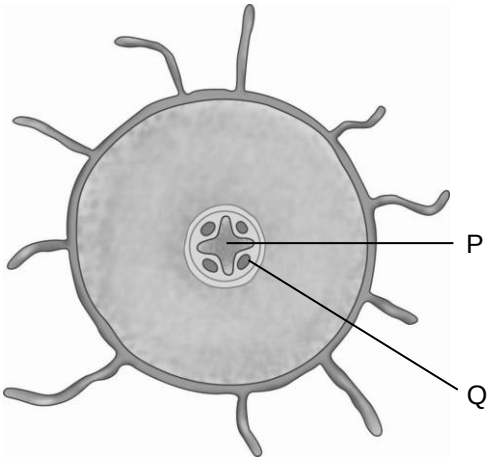
##A##

Skill assessed	Mark
Understanding basic concepts	1

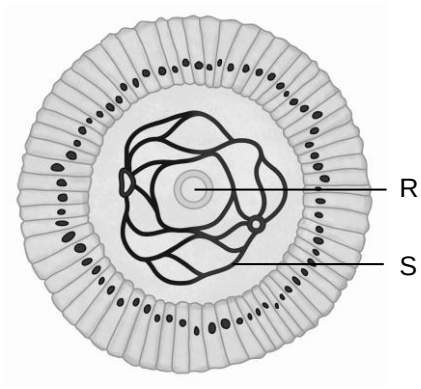
© 2019 Aristo Educational Press Ltd.

|!|BCEMB0100000018|!| ★ **Cross topic**

The following figures show the cross sections of a root and a villus of the small intestine.



Root



Villus of small intestine

Which of the following combinations correctly states the structures that are responsible for the uptake of water in plants and humans?

	Plants	***Humans***
A.	P	R
B.	Q	S
C.	P	S
D.	Q	R

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0100000021|!| ★

Which of the following is ***not*** a feature of a root hair cell for the uptake of water into the plant?

A. Root hair cell has a large surface area.
B. Root hair cell is long and fine.
C. Root hair cell has a thin cell wall.
D. Root hair cell has many mitochondria.

□

##D##

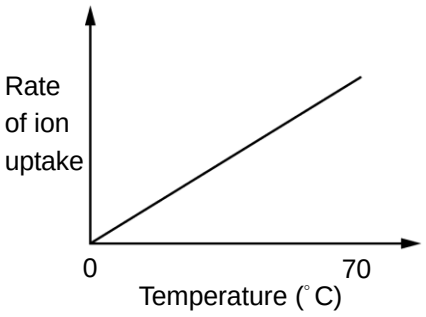
Skill assessed	Mark
Understanding basic concepts	1

© 2019 Aristo Educational Press Ltd.

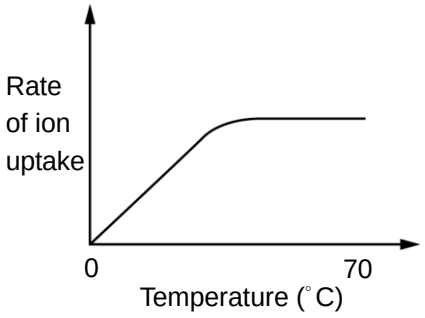
|!|BCEMA0100000022|!| ★★

Which of the following graphs ***best*** shows the effect of increasing temperature on the rate of uptake of mineral ions by root cells?

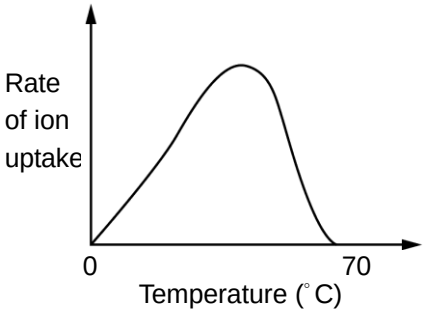
A.



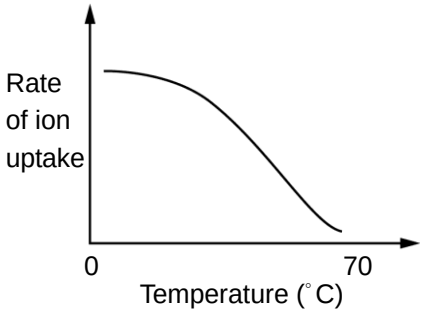
B.



C.



D.



□

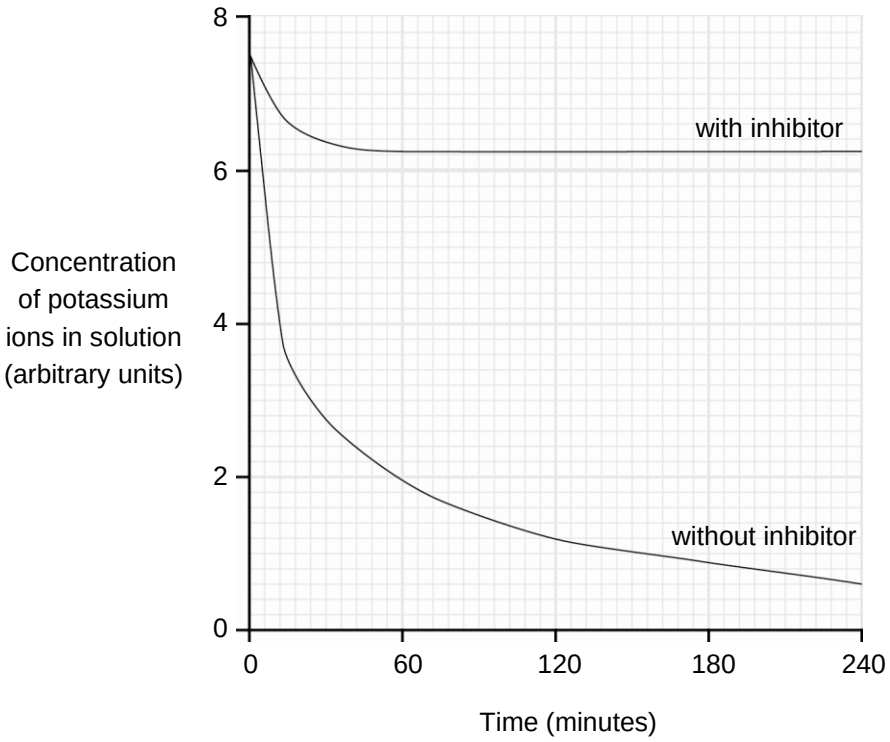
##C##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMA0100000023|!| ★★ **SI**

In an experiment, two root samples from a plant were placed in two beakers of aerated solution containing potassium ions. Potassium cyanide, a respiratory inhibitor, was added to one of the solutions. The concentrations of potassium ions in the two solutions were recorded at regular intervals. The following graph shows the results.



With reference to the graph, which of the following statements is ***incorrect***?

A. Between 0 to 30 minutes, the root absorbs potassium ions by diffusion in the presence of inhibitor.

B. After 60 minutes, there is no net movement of potassium ions between the root cells and the solution in the presence of inhibitor.

C. Between 0 to 30 minutes, the root absorbs potassium ions mainly by active transport in the absence of inhibitor.

D. After 60 minutes, there is a net movement of potassium ions down the concentration gradient from the solution into the root cells in the absence of inhibitor.

□

##D##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMC0100000024|!|

Which of the following leaf cells do ***not*** have chloroplasts?

(1) upper epidermal cell
(2) lower epidermal cell
(3) xylem cell
(4) guard cell

A. (1) and (2) only
B. (1), (2) and (3) only
C. (1), (2) and (4) only
D. (2), (3) and (4) only

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0100000026|!| **Cross topic**

Which of the following parts in the human body plays a similar function as the spongy mesophyll in a leaf?

A. red blood cells
B. air sacs in the lungs
C. villi in the ileum
D. heart chambers

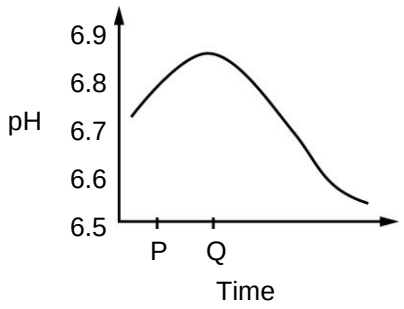
##B##

Skill assessed	Mark
Understanding basic concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMA0100000028|!| ★★

The graph below shows the changes in the pH of the water in a pond, which provides the habitat for many aquatic plants.



The increase in pH between time P and time Q is caused by

A. increased carbon dioxide concentration as the aquatic plants carry out photosynthesis.
B. decreased carbon dioxide concentration as the aquatic plants carry out photosynthesis.
C. increased carbon dioxide concentration as the aquatic plants carry out respiration.
D. decreased carbon dioxide concentration as the aquatic plants carry out respiration.

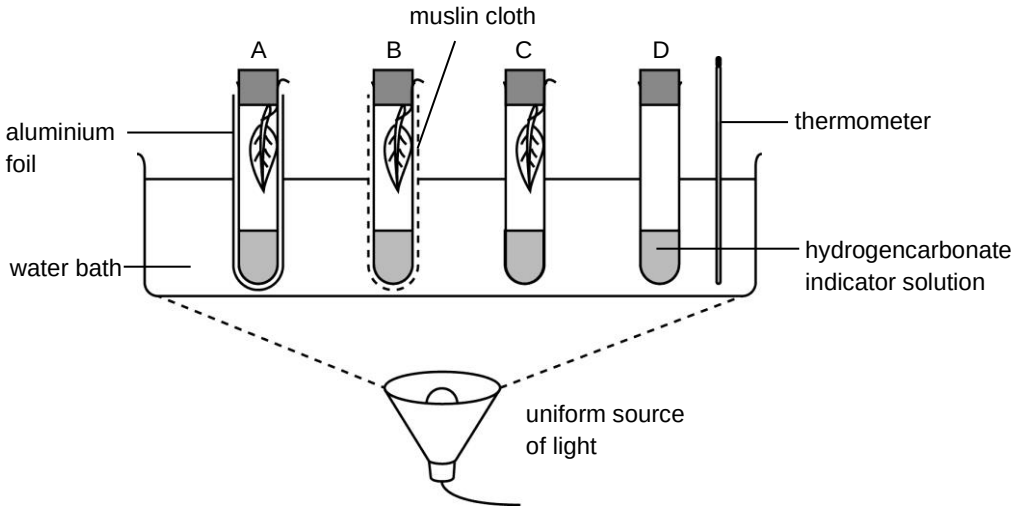
□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0100000029|!| **SI**

The following diagram shows an experimental set-up used to investigate the effect of light intensity on gas exchange in leaves.



© 2019 Aristo Educational Press Ltd.

Which of the following tubes will the indicator solution changes from red to yellow colour at the end of the experiment?

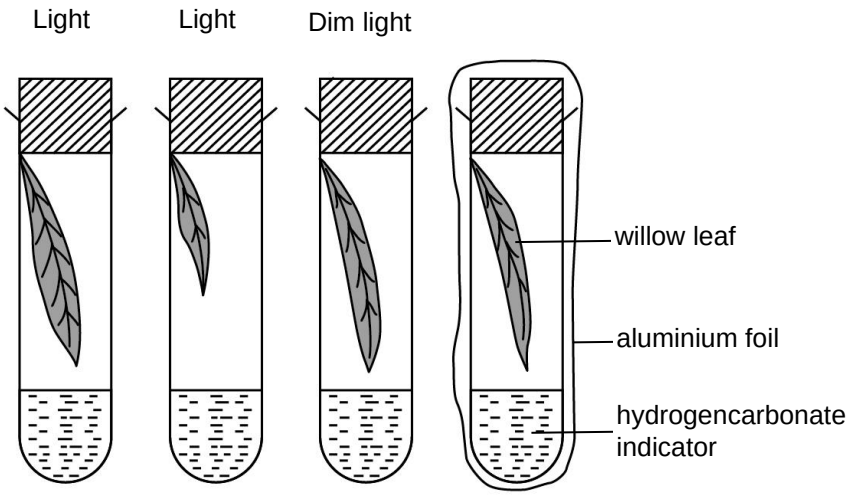
A. tube A
B. tube B
C. tube C
D. tube D

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0100000032|!| ★ **SI**

Equal volume of hydrogencarbonate indicator solution is added to each of the four boiling tubes. A willow leaf is also added to each of them.



How many independent variables is/are there in this investigation?

A. 0
B. 1
C. 2
D. 3

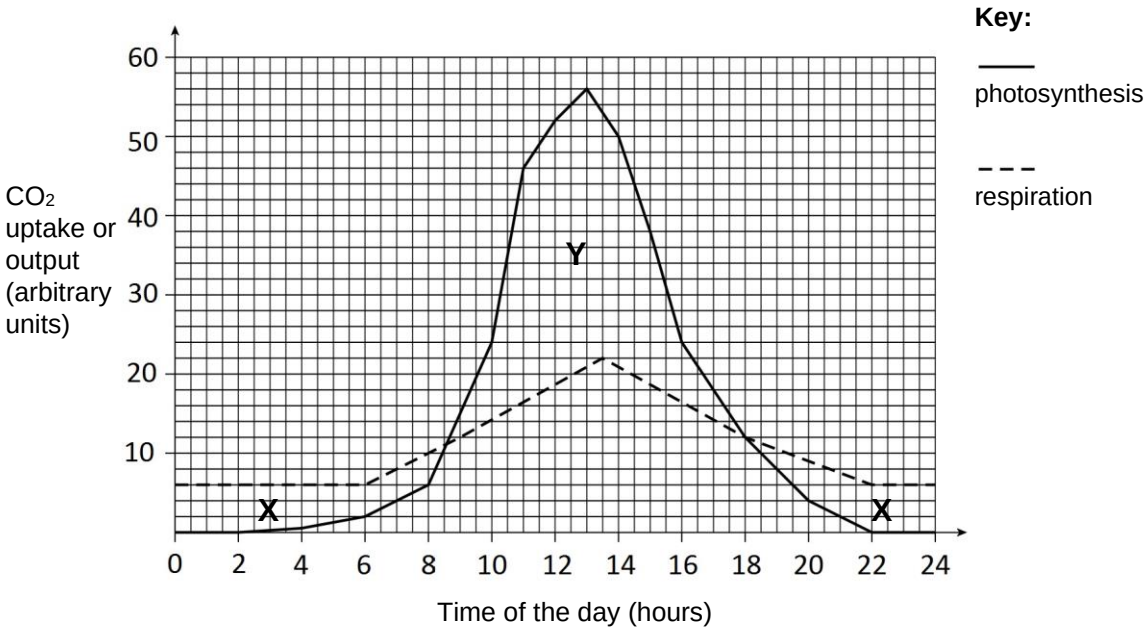
##C##

Skill assessed	Mark
Practical skills	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0100000033|!| ★

Directions: The following three questions refer to the graph below, which shows the rates of respiration and photosynthesis in a plant over a 24-hour period.



At which of the following times of the day is there the greatest net production of carbohydrate?

A. 0900 hours
B. 1300 hours
C. 1400 hours
D. 1800 hours

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0100000034|!| ★

With reference to the graph, which of the following statements is correct?

A. Both photosynthesis and respiration stop at midnight.
B. The rate of respiration is constant throughout the 24-hour period.
C. The rate of respiration is equal to that of photosynthesis at 0900 and 1800 hours.
D. The rate of respiration is not affected by light and temperature.

##C##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0100000035|!| ★

This is an actively growing plant because

A. the rate of photosynthesis is higher than that of respiration in the day time.
B. the rate of respiration is higher than that of photosynthesis at night.
C. area Y is greater than area X.
D. respiration occurs throughout the day to release energy.

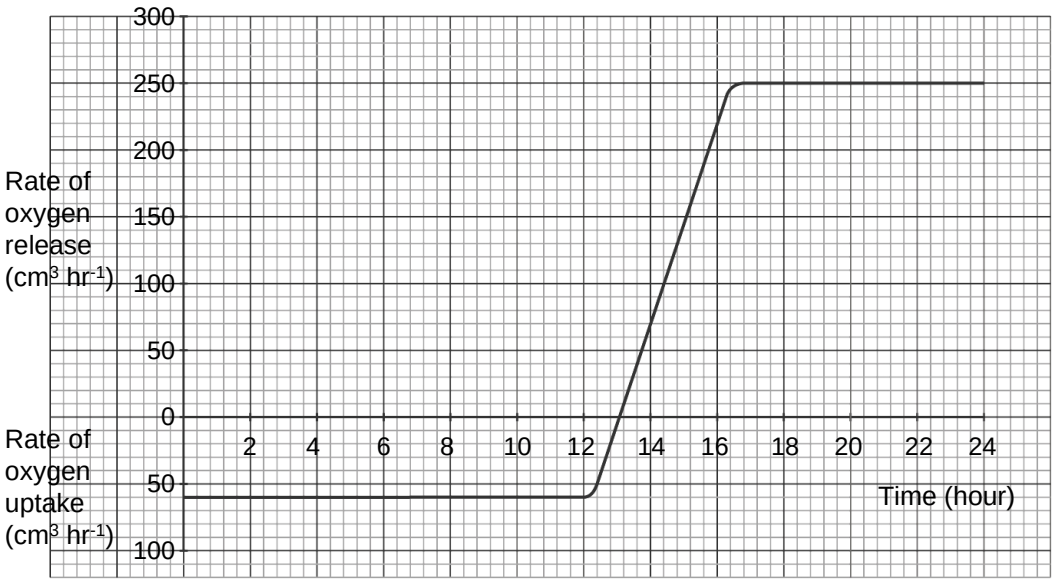
##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0100000036|!| ★★

Directions: The following two questions refer to the experiment below.

A group of plants were placed in the dark for 12 hours, followed by 12 hours in the light. The temperature was constant throughout the experiment. The rates of oxygen release and uptake in plants were shown in the graph below.



At 18 hours after the beginning of the experiment, what is the actual rate of photosynthesis of the plants in terms of the rate of oxygen release?

A. 0 $cm^3\ hr^{-1}$
B. 60 $cm^3\ hr^{-1}$
C. 250 $cm^3\ hr^{-1}$
D. 310 $cm^3\ hr^{-1}$

##D##

Skill assessed	Mark
Mathematical skills	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMA0100000037|!| ★★

Which of the following is the most accurate estimate of the total volume of oxygen used by the plants for respiration during the experiment?

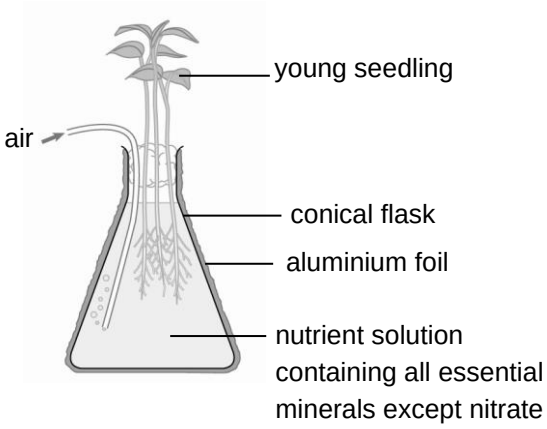
A. 310 cm^3
B. 720 cm^3
C. 1440 cm^3
D. 2000 cm^3

##C##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMB0100000038|!| ★ SI

Directions: The following three questions refer to the following experimental set-up, which is used to investigate the effect of nitrate on the growth of young seedlings.



What is the purpose of wrapping the conical flask with aluminium foil?

A. to prevent the seedlings from carrying out photosynthesis
B. to prevent the growth of algae in the nutrient solution
C. to provide insulation
D. to protect the conical flask from damage

##B##

Skill assessed	Mark
Practical skills	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0100000039|!| ★ **SI**

A control set-up is missing in this experiment. The control-set-up should be the same as the experimental set-up except that in the control set-up,

A. seedlings are removed.
B. the nutrient solution is replaced with distilled water.
C. the nutrient solution is replaced with distilled water with nitrate added.
D. the nutrient solution is replaced with another that contains all essential minerals.

##D##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0100000040|!| ★ **SI**

After one week, the seedlings show retarded growth. This is because nitrate is essential for

(1) making chlorophyll that are essential for carrying out photosynthesis.
(2) making nucleic acids that are involved in cell division.
(3) making proteins that are involved in the production of new cells

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

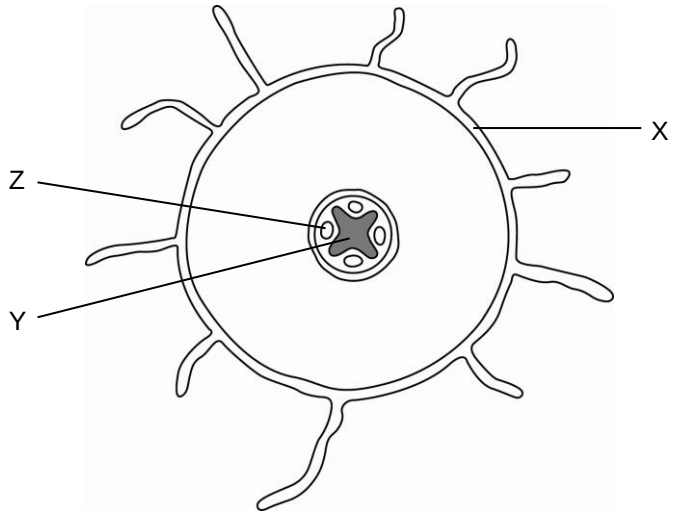
##D##

Skill assessed	Mark
Understanding basic concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0100000041|!| ★

Directions: The following two questions refer to the diagram below, which shows the transverse section of a dicotyledonous root.



Which of the following combinations correctly identifies the structures?

	X	***Y***	***Z***
A.	epidermis	xylem	phloem
B.	cuticle	phloem	xylem
C.	cuticle	xylem	phloem
D.	epidermis	phloem	xylem

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0100000042|!| ★

Which of the following statements about structure X are correct?

(1) Root hairs are developed from X.
(2) Mineral ions can be taken up into X by active transport.
(3) X is covered by a thin layer of cuticle.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

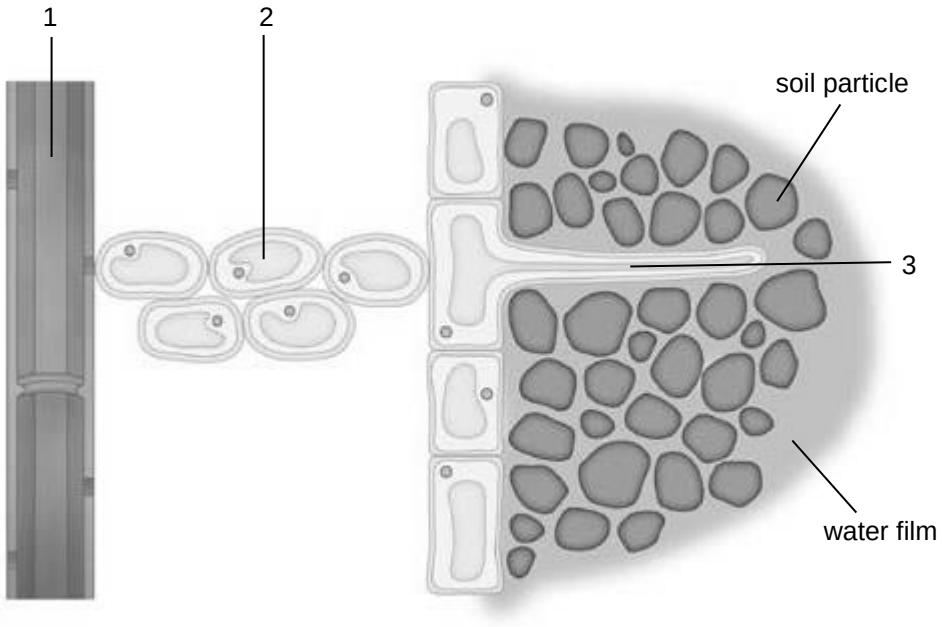
##A##

Skill assessed	Mark
Understanding basic concepts	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMA0100000043|!| ★★

Directions: The following two questions refer to the diagram below, which shows the transverse section of a certain part of a young root in the soil.



Which of the following cells is/are non-living?

A. 1 only
B. 1 and 2 only
C. 1 and 3 only
D. 2 and 3 only

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0100000044|!| ★★ **Cross topic**

Which of the following substances can pass into structure 3 mainly by simple diffusion?

(1) oxygen
(2) magnesium ion
(3) carbon dioxide

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

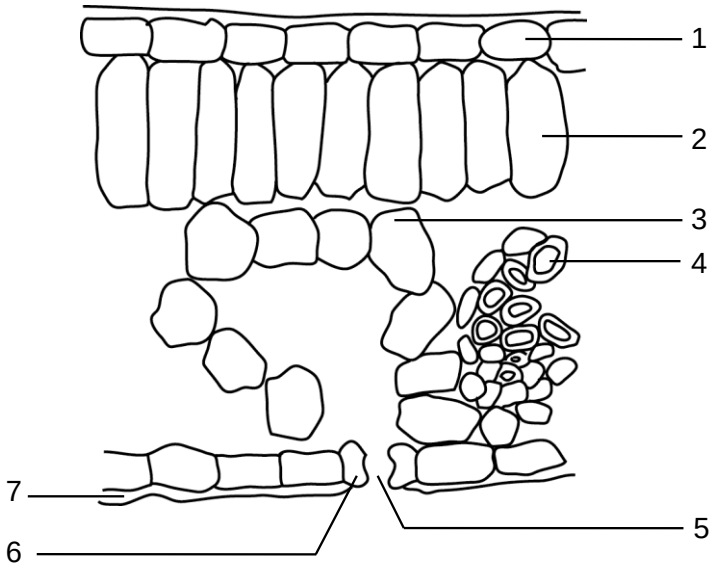
##B##

© 2019 Aristo Educational Press Ltd.

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0100000045|!| ★

Directions: The following two questions refer to the diagram below which shows a transverse section of a leaf.



Which of the following structures contain chloroplasts?

A. structures 2 and 3 only
B. structures 1, 2 and 3 only
C. structures 2, 3 and 5 only
D. structures 2, 3 and 6 only

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0100000046|!| ★

Which of the following pass out of the leaf through structure 5 during the daytime?

(1) oxygen
(2) carbon dioxide
(3) water vapour

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

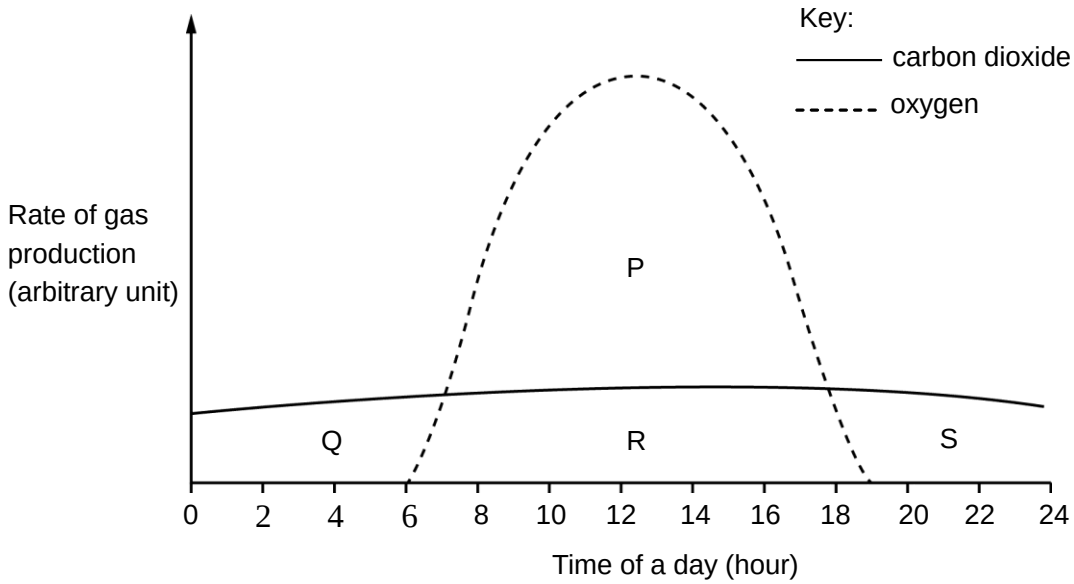
© 2019 Aristo Educational Press Ltd.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0100000047|!| ★★

Directions: The following two questions refer to the graph below, which shows the rates of oxygen and carbon dioxide production of a terrestrial plant during a day.



Which of the following shows the time at which there is no net exchange of oxygen and carbon dioxide between the leaves and the atmosphere?

A. 00:00
B. 06:00
C. 12:00
D. 18:00

□

##D##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMA0100000048|!| ★★

Photosynthesis is a food production process and respiration is a food consumption process. According to the different areas shown in the graph, which of the following can represent the net food production by the plant?

A. P
B. P + R
C. P – Q –S
D. P + R – Q –S

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0100000049|!| ★ SI

Directions: The following two questions refer to an experiment. In the experiment, the gas given off from the cut stem of a *Hydrilla* was analysed at different times of a day, with the temperature kept constant throughout the experiment. The results are shown in the following table.

Time	Weather	Percentage of carbon dioxide (%)	Percentage of oxygen (%)
7 a.m.	dull, cloudy	5	15
11 a.m.	slightly cloudy	0	30
2 p.m.	sunny	0	37
4 p.m.	sunny	0	43

Which of the following statements concerning the results is correct?

A. The percentage of oxygen in the gas given off at 7 a.m. is higher than that of the atmosphere.
B. High light intensity increases the rate of respiration.
C. The percentage of carbon dioxide in the gas given off at 7 a.m. is higher than that of the atmosphere.
D. High light intensity decreases the rate of photosynthesis.

##C##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.

|!|BCEMB0100000050|!| ★ **SI**

Which of the following processes take(s) place in *Hydrilla* from 2 p.m. to 4 p.m.?

(1) transpiration
(2) respiration
(3) photosynthesis

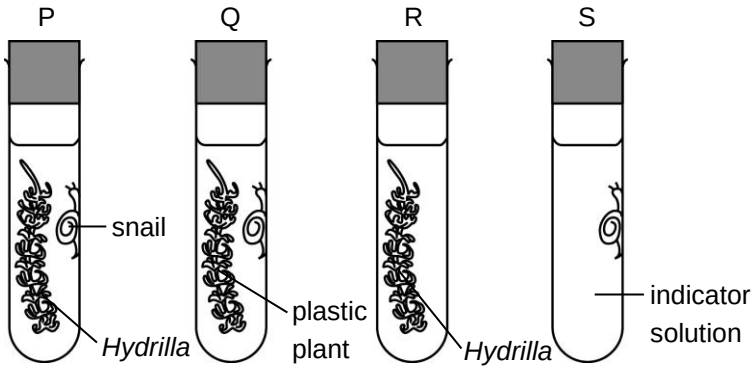
A. (3) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0100000051|!| ★ **SI**

Directions: The following two questions refer to the diagram below, which shows an experiment investigating how *Hydrilla* and snails interact with each other. The hydrogencarbonate indicator solution used was red in all the test tubes at the beginning. The tubes were illuminated with light of the same intensity.



The following table shows the colour changes of the hydrogencarbonate indicator solution under different concentrations of carbon dioxide (CO_2).

CO_2 concentration	Colour of indicator solution
< 0.03%	purple
~ 0.03%	red
> 0.03%	yellow

Which of the following combinations correctly shows the final colours of the indicator solutions in the test tubes after two hours?

© 2019 Aristo Educational Press Ltd.

	Tube P	*Tube Q*	*Tube R*	*Tube S*
A.	yellow	red	purple	yellow
B.	red	purple	yellow	purple
C.	red	yellow	purple	yellow
D.	purple	yellow	red	yellow

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0100000052|!| ★ **SI**

In another experiment, three tubes (T, U and V) were set up as in R. Tube T was then put in bright light, tube U in dim light, and tube V in the dark. Which of the following combinations correctly shows the final colours of the indicator solutions in the test tubes after two hours?

	Tube T	*Tube U*	*Tube V*
A.	purple	yellow	red
B.	purple	red	yellow
C.	yellow	purple	red
D.	red	yellow	purple

##B##

Skill assessed	Mark
Analysing data	1

© 2019 Aristo Educational Press Ltd.